6 Discrete-Time Fourier Transform

Objective

After completing this chapter, the user will be able to understand

1. Types to DTFT
2. Basic properties of DTFT
3. DTFT of various signals
4. Signal transmission through LTI system
5. Eigen value and Eigen function
6. System with periodic inputs

Introduction

1. We discussed the CTFT and developed the various characteristic and tools that makes the methods of Fourier analysis of such great value in analysing and understanding the properties of continuous time signals and system.
2. The discrete-time Fourier transform has essentially the same properties as the continuous time Fourier transform, and these properties play parallel roles in continuous time and discrete-time.
3. As with the continuous-time Fourier transform, the discrete-time Fourier transform is a complex-valued function whether or not the sequence is real-valued.

Discrete Time Fourier Transform (DTFT)

The discrete-time Fourier transform (DTFT) or the Fourier transform of a discrete-time periodic sequence x[n] is a representation of the sequence in terms of the complex exponential sequence $e^{j\omega n}$

$$\underset{\text{Discrete}}{x[n]} \overset{DTFT}{\leftrightarrow} X\left(e^{j\omega}\right) \underset{\text{continuous}}{or\ X\ \left(e^{j\Omega}\right)}$$

As in Fourier transform, X ($e^{j\omega}$) is also called spectrum and is a continuous function of the frequency parameter ω and is a periodic function of frequency with a period 2π.

Proof:

This derivation proceeds along a path analogous to the derivation of the CTFT from the CTFS. The DTFS representation of a periodic signal x(n) (with period N and frequency $\omega_0 = 2\pi/N$) can be written as

$$x\left(n\right) = \sum_{k=\left(N\right)} X_k e^{jk\omega_o n}$$

Substituting $X_k = 1/N \sum_{n=\{N\}} x\left(n\right)e^{-jk\omega_0 n}$ in the DTFS definition yields

$$x\left(n\right) = \sum_{k=\{N\}} \left(\frac{1}{N} \sum_{n=\{N\}} x\left(n\right)e^{-jk\omega_0 n} \right) e^{jk\omega_0 n}$$

or

$$x\left(n\right) = \sum_{k=\{N\}} \left(\sum_{n=\{N\}} x\left(n\right)e^{-jk\omega_0 n} \right) e^{jk\omega_0 n} \frac{\omega_0}{2\pi}$$

Since the inner summation is over any arbitrary range of n of width N, let the range be $-\left(N/2\right) \le n \le \left(N/2\right) - 1$ for N even or $-\left(N-1/2\right) \le n \le \left(N-1/2\right)$ for N odd. The outer summation is over any arbitrary range of k of width N, so let its range be $k_0 \le k \le \left(k_0 + N - 1\right)$. Then

$$x\left(n\right) = \sum_{k=k_0}^{k_0+N-1} \left(\sum_{n=-\frac{N}{2}}^{\frac{N}{2}-1} x(n)e^{-jk\omega_0 n} \right)$$

$e^{jk\omega_0 n\frac{\omega_0}{2\pi}}$, *for N even* (6.1)

Or

$$x\left(n\right) = \sum_{k=k_0}^{k_0+N-1} \left(\sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} x(n)e^{-jk\omega_0 n} \right)$$

$e^{jk\omega_0 n\frac{\omega_0}{2\pi}}$, *for N odd* (6.2)

Now let the fundamental period $N \to \infty$. In that limit $w_0/2\pi \to dw/2\pi$, and $kw_0 \to w$. The inner summation covers and infinite range. The outer summation approaches an integral in $w = k\omega_0$ covering a range of

$$k_0 \le k \le \left(k_0 + N - 1\right)$$
$$k_0 \le k \le \left(k_0 + N\right)$$

$$k_0 \le \frac{w}{w_0} \le (k_0 + N)$$

$$k_0 w_0 < w < (k_0 w_0 + N w_0)$$

$$k_0 w_0 < w < (k_0 w_0 + 2\pi)$$

Therefore, Eqs (6.1) and (6.2) both become

$$x(n) = \frac{1}{2\pi} \int_{2\pi} \left(\sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \right) e^{-j\omega n} d\omega \qquad (6.3)$$

Where $\int_{2\pi}$ indicates integration over any continuous interval of 2π.. The inner summation of the above equation is a function of only ω and not n. Denoting the summation by $X(e^{jw})$, we can write Eq. (6.3) as

$$x(n) = \frac{1}{2\pi} \int_{2\pi} X(e^{jw}) e^{jwn} \, dw \qquad (6.4)$$

Where $X(e^{jw}) = \sum_{n=-\infty}^{\infty} x(n) e^{-jwn}$ (6.5)

Equation (6.4) is usually referred to as the synthesis equation eq. (6.5) is referred to as the analysis equation. We call $X(e^{jw})$ the discrete-time Fourier transform (DTFT) of $x(n)$, and $x(n)$ the inverse discrete-time Fourier transform (IDTFT) of $X(e^{jw})$. The nomenclature can be represented as

$X(e^{jw}) = DTFT[x(n)] = f[x(n)]$ and

$x(n) = IDTFT[X(e^{jw})] = F^{-1}[X(e^{jw})]$

$X(e^{jw}) = \sum_{n=-\infty}^{\infty} x[n] e^{-jwn}$ Analysis equation

The infinite series in equation 1 may be converges or may not. $x[n]$ is absolutely summable.

When $\sum_{n=-\infty}^{\infty} |x(n)| < \infty$

An absolutely summable sequence always has a finite energy, but a finite-energy sequence is not necessarily to be absolutely summable.

1. Find DTFT of $x[n] = a^n u[n]\ ; 0 < a < 1$

Solution: $X(e^{jw}) = \sum_{n=-\infty}^{\infty} x[n] e^{-jwn}$

$$= \sum_{n=-\infty}^{\infty} a^n u[n] e^{-jwn} = \sum_{n=0}^{\infty} a^n e^{-jwn}$$

$$X(e^{jw}) = \sum_{n=0}^{\infty} (ae^{-jw})^n$$

$$= 1 + ae^{-jw} + (ae^{-jw})^2 + \cdots\cdots$$

$$X(e^{jw}) = \frac{1}{1 - ae^{-jw}} = \frac{1}{1 - a(\cos\omega - j\sin\omega)}$$

$$= \frac{1}{1 - a\cos\omega + ja\sin\omega}$$

$$|X(e^{jw})| = \frac{1}{\sqrt{(1 - a\cos\omega)^2 + a^2 \sin^2 \omega}}$$

$$= \frac{1}{\sqrt{1 - 2a\cos\omega + a^2}}$$

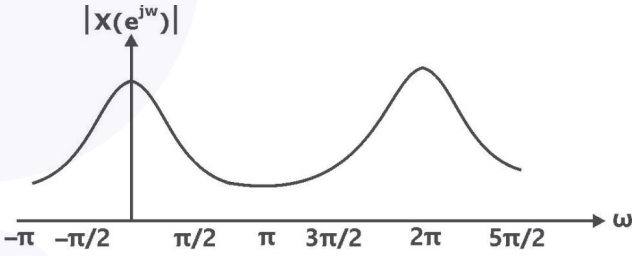


Fig. 6.1

2. Find DTFT of $x[n] = 2^n u[-n]$

Solution: $X(e^{jw}) = \sum_{n=-\infty}^{\infty} x[n] e^{-jwn} = \sum_{m=-\infty}^{0} 2^n e^{-jwn}$

$$= \sum_{m=0}^{\infty} \left\{ \left(\frac{1}{2}\right) e^{jw} \right\}^m$$

$$X(e^{j\omega}) = 1 + \frac{1}{2} e^{j\omega} + \left(\frac{1}{2} e^{j\omega}\right)^2 + \cdots\cdots$$

$$X(e^{j\omega}) = \frac{1}{\left(1 - \frac{1}{2} e^{j\omega}\right)}$$

3. Find DTFT of $x[n] = \left(\frac{1}{4}\right) u[n+2]$

Solution: $x(e^{j\omega}) = \sum_{n=-2}^{\infty} \left(\frac{1}{4}\right)^n e^{-j\omega n} = \left(\frac{1}{4}\right)^{-2} e^{-j\omega(-2)}$

$$+\left(\frac{1}{4}\right)^{-1} e^{-j\omega(-1)} + \sum_{m=0}^{\infty}\left(\frac{1}{4}\right)^{n} e^{-j\omega n}$$

$$x\left(e^{j\omega}\right) = 16e^{j2\omega} + 4e^{j\omega} + \frac{1}{1-\frac{1}{4}e^{-j\omega}} = \frac{16e^{j2\omega}}{1-\frac{1}{4}e^{-j\omega}}$$

Or

Put m = n + 2

$$\therefore X\left(e^{j\omega}\right) = \sum_{m=0}^{\infty}\left(\frac{1}{4}\right)^{m-2} e^{-j\omega(m-2)}$$

$$= 16e^{j2\omega}\sum_{m=0}^{\infty}\left(\frac{1}{4}\right)^{m} e^{-jwm} = \frac{16e^{j2\omega}}{1-\frac{1}{4}e^{-j\omega}}$$

4. Find DTFT of

$$y[n] = \sum_{n=-3}^{3} x[n]e^{-j\omega n} \quad if \quad x[n] =$$

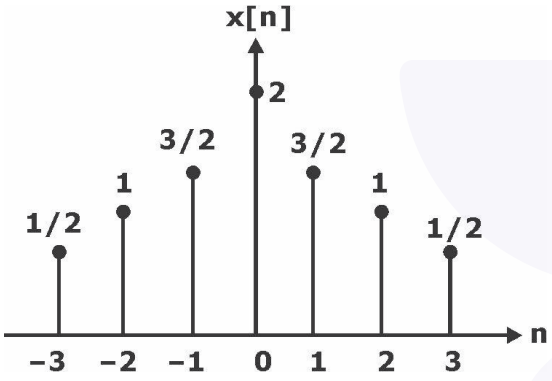


Fig. 6.2

Solution: $Y\left(e^{j\omega}\right) = \frac{1}{2}e^{j3\omega} + 1e^{j\omega 2} + \frac{3}{2}e^{j\omega} + 2 + \frac{3}{2}e^{-j\omega} + e^{-j2\omega} + \frac{1}{2}e^{-j3\omega}$

$$Y\left(e^{j\omega}\right) = \cos(3\omega) + 2\cos(2\omega) + 3\cos\omega + 2$$

5. Find the inverse Fourier transform of

$$X\left(e^{j\omega}\right) = \sum_{m=-\infty}^{\infty} 2\pi\delta\left(\omega - \omega_0 - 2\pi m\right)$$

Which is illustrated in the fig.?

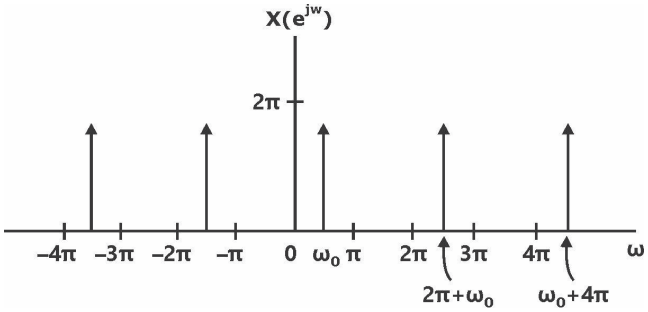


Fig. 6.3

Solution:

From fig., we can observe that

$$X\left(e^{j\omega}\right) = 2\pi\delta\left(\omega - \omega_0\right), \quad 0 \le \omega \le 2\pi$$

Now, from Eq. (6.4), we have

$$x(n) = \frac{1}{2\pi}\int_{2\pi} X\left(e^{j\omega}\right)e^{j\omega n}d\omega$$

Where $\int_{2\pi}$ indicates integration over any continuous interval of 2π. Therefore,

$$x(n) = \frac{1}{2\pi}\int_{0}^{2\pi} X\left(e^{j\omega}\right)e^{j\omega n}d\omega$$

$$= \frac{1}{2\pi}\int_{0}^{2\pi} 2\pi\delta\left(\omega - \omega_0\right)e^{j\omega n}d\omega$$

$$x(n) = e^{j\omega n}\Big|_{\omega=\omega_0} = e^{j\omega_0 n}$$

$$e^{j\omega_0 n} \longleftrightarrow \sum_{m=-\infty}^{\infty} 2\pi\delta\left(\omega - \omega_0 - 2\pi m\right) \quad (6.6)$$

Properties of DTFT

Linearity:

$$If \; x_1[n] \xleftrightarrow{DTFT} X_1\left(e^{j\omega}\right) \; and \; x_2[n] \xleftrightarrow{DTFT} X_2\left(e^{j\omega}\right)$$

$$Then \; \alpha x_1[n] + \beta x_2[n] \xleftrightarrow{DTFT} \alpha X_1\left(e^{j\omega}\right) + \beta X_2\left(e^{j\omega}\right)$$

Time shifting:

$$If \; x[n] \xleftrightarrow{DTFT} X\left(e^{j\omega}\right)$$

$$Then \; x[n-n_0] \xleftrightarrow{DTFT} e^{-j\omega n_0}X\left(e^{j\omega}\right)$$

Proof:

$$DTFT\left\{x\left(n-n_0\right)\right\} = \sum_{n=-\infty}^{\infty} x[n-n_0]e^{-j\omega n}$$

$$put \; n - n_0 = m \Rightarrow n = m + n_0$$

$$DTFT\left\{x\left(n-n_0\right)\right\} = \sum_{n=-\infty}^{\infty} x[m]e^{-j\omega(m+n_0)}$$

$$= e^{-j\omega n_0}\sum_{n=-\infty}^{\infty} x[m]e^{-j\omega n}$$

$$= e^{-j\omega n_0}X\left(e^{j\omega}\right)$$

Frequency shifting:

$If\ x[n] \xleftrightarrow{DTFT} X(e^{j\omega})$

$Then\ x[n]e^{j\omega n_0} \xleftrightarrow{DTFT} X(e^{j(\omega-\omega_0)})$

Proof:

$$DTFT\{x[n]e^{j\omega_0 n}\} = \sum_{n=-\infty}^{\infty} x[n]e^{j\omega_0 n}\, e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} x[n]e^{-j(\omega-\omega_0)n}$$

$$= X(e^{j(\omega-\omega_0)})$$

Time scaling:

$If\ x[n] \xleftrightarrow{DTFT} X(e^{j\omega})$

$Then\ x\left[\frac{n}{k}\right] \xleftrightarrow{DTFT} X(e^{j\omega k})$

Proof:

$$DTFT\left\{x\left[\frac{n}{k}\right]\right\} = \sum_{n=-\infty}^{\infty} x\left[\frac{n}{k}\right]e^{-j\omega n}$$

Put n/k = m

$$DTFT\left\{x\left[\frac{n}{k}\right]\right\} = \sum_{n=-\infty}^{\infty} x\left[\frac{n}{k}\right]e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} x[m]e^{-j\omega mk} = X(e^{j\omega k})$$

Time reversal:

$If\ x[n] \xleftrightarrow{DTFT} X(e^{j\omega})$

$Then\ x[-n] \xleftrightarrow{DTFT} X(e^{-j\omega})$

Proof:

$$DTFT\{x[-n]\} = \sum_{n=-\infty}^{\infty} x[-n]e^{-j\omega n}$$

Put -n = m

$$DTFT\{x[-n]\}$$

$$= \sum_{m=\infty}^{-\infty} x[m]e^{-j\omega(-m)}$$

$$= \sum_{m=-\infty}^{\infty} x[m]e^{-j(-\omega)m} = X(e^{-j\omega})$$

Conjugate:

$If\ x[n] \xleftrightarrow{DTFT} X(e^{j\omega})$

$Then\ x^*[n] \xleftrightarrow{DTFT} X^*(e^{-j\omega})$

Proof:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n]e^{-j\omega n}$$

Replace 'ω' by '-ω'

$$X(e^{-j\omega}) = \sum_{n=-\infty}^{+\infty} x[n]e^{j\omega n}$$

Conjugate both sides

$$X^*(e^{-j\omega}) = \sum_{n=-\infty}^{+\infty} x^*[n]e^{-j\omega n}$$

Frequency differentiation:

$If\ x[n] \xleftrightarrow{DTFT} x(e^{j\omega})$

$Then\ nx[n] \xleftrightarrow{DTFT} j\frac{d}{d\omega}X(e^{j\omega})$

Proof:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n]e^{-j\omega n}$$

Differentiating both sides w.r.t. 'ω'

$$\frac{d}{d\omega}X(e^{j\omega}) = \frac{1}{j}\sum_{n=-\infty}^{+\infty}\{nx[n]\}(e^{-j\omega n})$$

$$\therefore nx[n] \xleftrightarrow{DTFT} j\frac{d}{d\omega}X(e^{j\omega})$$

Convolution in n-domain:

$If\ x_1[n] \xleftrightarrow{DTFT} X_1(e^{j\omega})\ and$

$x_2[n] \xleftrightarrow{DTFT} X_2(e^{j\omega})$

$Then\ x_1[n]*x_2[n] \xleftrightarrow{DTFT} X_1(e^{j\omega}) \times X_2(e^{j\omega})$

Proof:

$$DTFT\{x_1[n] * x_2[n]\}$$

$$= \sum_{n=-\infty}^{\infty}\{x_1[n] * x_2[n]\}e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty}\sum_{m=-\infty}^{\infty} x_1[m]x_2[n-m]e^{-j\omega n}$$

Put P = n - m => n = p + m

$$DTFT\{x_1[n] * x_2[n]\}$$

$$= \sum_{p=-\infty}^{\infty}\sum_{m=-\infty}^{\infty} x_1[m]x_2[p]e^{-j\omega(p+m)}$$

$$= \sum_{m=-\infty}^{\infty} x_1[m]e^{-j\omega m}\sum_{p=-\infty}^{\infty} x_2[p]e^{-j\omega p}$$

$$DTFT\{x_1[n] * x_2[n]\} = X_1(e^{j\omega})X_2(e^{j\omega})$$

Windowing or modulation property:

$$If\ x_1[n] \xleftrightarrow{DTFT} X_1(e^{j\omega})$$
$$and\ x_2[n] \xleftrightarrow{DTFT} X_2(e^{j\omega})$$

$$Then\ x_1[n]x_2[n] \xleftrightarrow{DTFT} \frac{1}{2\pi}\begin{pmatrix} X_1(e^{j\omega}) * \\ X_2(e^{j\omega}) \end{pmatrix}$$

$$or\ \frac{1}{2\pi}\int_{-\pi}^{\pi} X_1(e^{jp})X_2(e^{j(\omega-p)})dp$$

Proof:

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} X_1(e^{jp})X_2(e^{j(\omega-p)})dp$$
$$= \frac{1}{2\pi}\int_{-\pi}^{\pi}\left(\sum_{m=-\infty}^{\infty} x_1[m]e^{-jpm}\right)\left(\sum_{n=-\infty}^{\infty} x_2[n]e^{-j(\omega-p)n}\right)dp$$
$$= \frac{1}{2\pi}\sum_{m=-\infty}^{\infty}\sum_{n=-\infty}^{\infty} x_1[m]x_2[n]\int_{-\pi}^{\pi}\left(e^{-jpm}e^{-j(\omega-p)n}\right)dp$$
$$= \frac{1}{2\pi}\sum_{m=-\infty}^{\infty}\sum_{n=-\infty}^{\infty} x_1[m]x_2[n]e^{-j\omega n}\int_{-\pi}^{\pi}\left(e^{-jp(n-m)}\right)dp$$

$$\int_{-\pi}^{\pi}\left(e^{-jp(n-m)}\right)dp \neq 0\ iff\ m = n$$

$$\frac{1}{2\pi}\sum_{n=-\infty}^{\infty} x_1[n]x_2[n]e^{-j\omega n} \times 2\pi$$
$$= \sum_{n=-\infty}^{\infty} x_1[n]x_2[n]e^{-j\omega n}$$
$$= DTFT\{x_1[n]x_2[n]\}$$

Periodicity:

$$If\ x[n] \xleftrightarrow{DTFT} X(e^{j\omega})$$
$$Then\ x\{e^{j(\omega+2\pi k)}\} = X(e^{j\omega})$$

Proof:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\ e^{-j\omega n}$$

$$Replace\ '\omega'\ by\ '\omega + 2\pi k'$$

$$X(e^{j(\omega+2\pi k)}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j(\omega+2\pi k)} = X(e^{j\omega})$$

First difference:

$$If\ x[n] \xleftrightarrow{DTFT} X(e^{j\omega})$$

$$Then\ x[n] - x[n-1] \xleftrightarrow{DTFT} (1 - e^{-j\omega})x(e^{j\omega})$$

Proof:

$$DTFT\{x[n] - x[n-1]\} = X(e^{j\omega})$$
$$-X(e^{j\omega})e^{-j\omega} = (1 - e^{-j\omega})X(e^{j\omega})$$

Parseval's theorem:

$$If\ x[n] \xleftrightarrow{DTFT} X(e^{j\omega})$$

$$Then\ \sum_{n=-\infty}^{\infty}|x[n]|^2 = \frac{1}{2\pi}\int_{<2\pi>}|X(e^{j\omega})|^2 d\omega$$

Proof:

$$x[n] = \frac{1}{2\pi}\int_{<2\pi>} X(e^{j\omega})e^{j\omega n}d\omega$$

$$x^*[n] = \frac{1}{2\pi}\int_{<2\pi>} X^*(e^{j\omega})e^{-j\omega n}d\omega$$

LHS =

$$\sum_{n=-\infty}^{\infty}|x[n]|^2 = \sum_{n=-\infty}^{\infty} x[n]x^*[n]$$
$$= \sum_{n=-\infty}^{\infty} x[n]\frac{1}{2\pi}\int_{<2\pi>} X^*(e^{j\omega})e^{-j\omega n}d\omega$$

Adjusting summation and integration;

$$\textbf{LHS} = \frac{1}{2\pi}\int_{<2\pi>} X^*(e^{j\omega})\sum_{n=-\infty}^{\infty}|x[n]|e^{-j\omega n}d\omega$$
$$= \frac{1}{2\pi}\int_{<2\pi>} X^*(e^{j\omega})X(e^{j\omega})$$
$$= \frac{1}{2\pi}\int_{<2\pi>}|X(e^{j\omega})|^2 d\omega$$

Fourier Transform of Periodic Signals

1. We can construct the Fourier transform of a periodic signal directly from its Fourier series representation.
2. The resulting transform consists of a train of impulses in the frequency domain, with the area of the impulses proportional to the Fourier series coefficients.

   Consider the periodic signal x(n) with period N and fundamental frequency $\omega_0 = 2\pi / N$. The signal x(n) has the discrete-time Fourier series representation

   $$x(n) = \sum_{k=\langle N\rangle} X_k e^{jk\omega_0 n}$$

Taking the Fourier transform on both sides yields

$$F\left|x(n)\right| = F\left[\sum_{k=\langle N\rangle} X_k e^{jk\omega_0 n}\right]$$

$$X\left(e^{j\omega}\right) = \sum_{k=\langle N\rangle} X_k F\left[e^{jk\omega_0 n}\right]$$

Using Eq. 6.6 we have

$$F\left[e^{jk\omega_0 n}\right] = 2\pi\delta\left(\omega - k\omega_0\right) \quad 0 \le \omega < 2\pi$$

So that

$$X\left(e^{j\omega}\right) = \sum_{k=\langle N\rangle} X_k 2\pi\delta\left(\omega - k\omega_0\right), \quad 0 \le \omega < 2\pi$$

$$X\left(e^{j\omega}\right) = 2\pi\sum_{k=\langle N\rangle} X_k \delta\left(\omega - k\omega_0\right), \quad 0 \le \omega < 2\pi$$

Since the DTFT is periodic with period 2π, it follows that $X\left(e^{j\omega}\right)$ consists a set of N impulses of strength $2\pi X_k$, k = 0,1,2,..., N-1, repeated at intervals of $N\omega_0 = 2\pi$. Thus, $X(e^{j\omega})$ can be compactly written as

$$X\left(e^{j\omega}\right) = 2\pi\sum_{k=-\infty}^{\infty} X_k \delta\left(\omega - k\omega_0\right)$$

Thus, the Fourier transform of a periodic signal is simply an impulse train with impulses located at $\omega = k\omega_0$, each of which has a strength $2\pi X_k$, and all impulses are separated from each other by ω_0.

Signal Transmission through LTI System

If x(n) and y(n) are the input and output of an LTI system with impulse response h(n).

$$y(n) = x(n) * h(n)$$

Application of the time convolution property to the above equation yields [assuming that both x(n) and h(n) are Fourier transformable]

$$F\left[y(n)\right] = F\left[x(n)\right] * F\left[h(n)\right]$$

$$Y\left(e^{j\omega}\right) = X\left(e^{j\omega}\right)H\left(e^{j\omega}\right) \qquad (6.7)$$

Or $$H\left(e^{j\omega}\right) = \frac{Y\left(e^{j\omega}\right)}{X\left(e^{j\omega}\right)} \qquad (6.8)$$

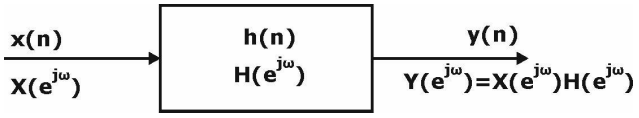


Fig. 6.4

The use of the convolution property for LTI systems is demonstrated in fig. equation (6.8) can be expressed in the polar form as

$$\left|Y\left(e^{j\omega}\right)\right| e^{j\angle Y\left(e^{j\omega}\right)} = \left[\left|X\left(e^{j\omega}\right)\right| e^{j\angle X\left(e^{j\omega}\right)}\right] \left[\left|H\left(e^{j\omega}\right)\right| e^{j\angle H\left(e^{j\omega}\right)}\right]$$

$$\left|Y\left(e^{j\omega}\right)\right| e^{j\angle Y\left(e^{j\omega}\right)} = \left|X\left(e^{j\omega}\right)\right|\left|H\left(e^{j\omega}\right)\right| e^{j\left|\angle X\left(e^{j\omega}\right)+\angle H\left(e^{j\omega}\right)\right|}$$

Therefore, $$\left|Y\left(e^{j\omega}\right)\right| = \left|X\left(e^{j\omega}\right)\right|\left|H\left(e^{j\omega}\right)\right|$$

And $$\angle Y\left(e^{j\omega}\right) = \angle X\left(e^{j\omega}\right) + \angle H\left(e^{j\omega}\right) \qquad (6.9)$$

During transmission, the input signal magnitude spectrum $\left|X\left(e^{j\omega}\right)\right|$ is changed to $\left|X\left(e^{j\omega}\right)\right|\left|H\left(e^{j\omega}\right)\right|$. Similarly, the input signal phase spectrum $\angle X\left(e^{j\omega}\right)$ is changed to $\angle X\left(e^{j\omega}\right) + \angle H\left(e^{j\omega}\right)$. $\left|H\left(e^{j\omega}\right)\right|$ is the magnitude response and $\angle H\left(e^{j\omega}\right)$ is the phase response of the system. $H\left(e^{j\omega}\right)$ is called the frequency response of the system.

6. Find the Fourier transform of the periodic signal.

$$x(n) = \cos\left(\omega_0 n\right)$$

With $\omega_0 = 2\pi/5$

Solution:

Consider the given signal

$$x(n) = \cos\left(\omega_0 n\right) = \frac{1}{2}e^{j\omega_0 n} + \frac{1}{2}e^{-j\omega_0 n}$$

We know that

$$e^{j\omega_0 n} \longleftrightarrow \sum_{m=-\infty}^{\infty} 2\pi\delta\left(\omega - \omega_0 - 2\pi m\right)$$

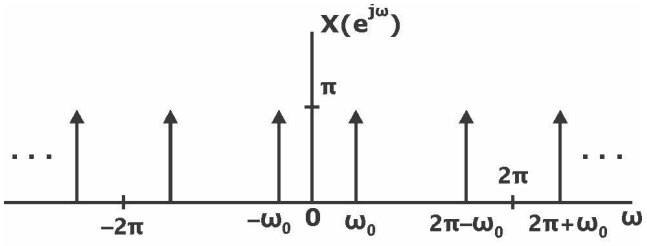
Therefore,

$$X\left(e^{j\omega}\right) = \frac{1}{2}F\left[e^{j\omega_0 n}\right] + \frac{1}{2}F\left[e^{-j\omega_0 n}\right]$$

$$X\left(e^{j\omega}\right) = \frac{1}{2}\sum_{m=-\infty}^{\infty} 2\pi\delta\left(\omega - \omega_0 - 2\pi m\right) + \frac{1}{2}\sum_{m=-\infty}^{\infty} 2\pi\delta\left(\omega + \omega_0 - 2\pi m\right)$$

$$X\left(e^{j\omega}\right)=\frac{1}{2}\sum_{m=-\infty}^{\infty}2\pi\delta\left(\omega-\frac{2\pi}{5}-2\pi m\right)+\frac{1}{2}\sum_{m=-\infty}^{\infty}2\pi\delta\left(\omega-\frac{2\pi}{5}-2\pi m\right)$$

The DTFT $X\left(e^{j\omega}\right)$ is illustrated in the fig.



DTFT fo x(n) = cos(ω_0n)

Fig. 6.5

7. Find the DTFT of $na^nu[n]$.

Solution: Let $a^nu[n]=x[n]$ such that

$$y[n]=nx[n]$$

$$X\left(e^{j\omega}\right)=\frac{1}{1-ae^{-j\omega}}$$

$$y[n]=nx[n]$$

Applying DTFT

$$Y\left(e^{j\omega}\right)=j\frac{d}{d\omega}\left\{\frac{1}{1-ae^{-j\omega}}\right\}=\frac{ae^{-j\omega}}{\left(1-ae^{-j\omega}\right)^2}$$

8. Find the DTFT of

a) $x[n]=4(0.5)^{n+3}u[n]$

b) $y[n]=n(0.4)^{2n}u[n]$

Solution: a) $x[n]=4(0.5)^{n+3}u[n]$

$$=4(0.5)^3(0.5)^nu(n)$$

$$=(0.5)\left\{(0.5)^nu(n)\right\}$$

$$x\left(e^{j\omega}\right)=\frac{0.5}{1-0.5e^{-j\omega}}=\frac{0.5}{1-0.5e^{-j\omega}}$$

b) $y[n]=n(0.4)^{2n}u[n]=n(0.16)^nu[n]$

$$Y\left(e^{j\omega}\right)=j\frac{d}{d\omega}\left\{\frac{1}{1-0.16e^{-j\omega}}\right\}=\frac{0.16e^{-j\omega}}{\left(1-0.16e^{-j\omega}\right)^2}$$

Linear and Non-linear Phase

1. Consider a discrete-time LTI system with impulse response h(n) and frequency response H ($e^{j\omega}$).
2. Let a signal x(n) with Fourier transform X($e^{j\omega}$) be applied to the input of the system.
3. Let a signal y(n) with Fourier transform Y($e^{j\omega}$) denote the output of the system.
4. In distortedness transmission, the input x(n) and the output y(n) satisfy the condition.



Fig. 6.6

$y(n) = Gx(n - n_d)$

Where the constant G accounts for a change in amplitude and the constant n_d accounts for a delay in transmission. Taking the Fourier transform of the above equation yields

$$Y\left(e^{j\omega}\right)=GX\left(e^{j\omega}\right)e^{-j\omega n_d}$$

$$\frac{Y\left(e^{j\omega}\right)}{X\left(e^{j\omega}\right)}=H\left(e^{j\omega}\right)=Ge^{-j\omega n_d}$$

This is the frequency response required of a system for distortion less transmission. From this equation, it follows that

a) The magnitude response $\left|H\left(e^{j\omega}\right)\right|$ must be a constant, i.e., we must have

$$\left|H\left(e^{j\omega}\right)\right|=G$$

For some constant G

b) The phase response $\angle H\left(e^{j\omega}\right)$ must be a linear function of ω with slope $-n_d$ and intercept zero, i.e., we must have

$$\angle H\left(e^{j\omega}\right)=-\omega n_d$$

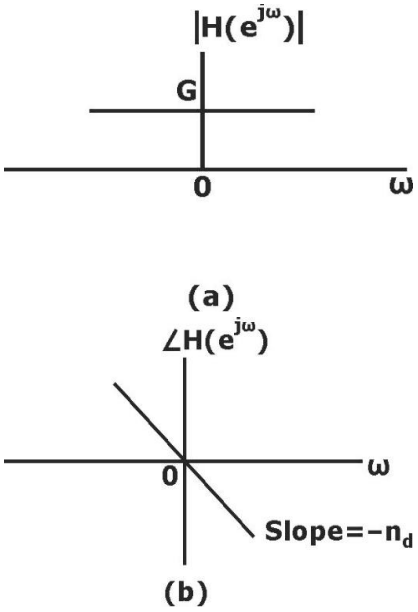


Fig.6.7 (a) Magnitude response and (b) phase response of a system for distortion less transmission

Phase delay:

The time delay experienced by a single-frequency signal (i.e., a sinusoidal signal) when the signal passes through a system is referred to as the system phase delay.

Now, let us assume that the input signal is the single-frequency signal

$$x(n) = A\cos(\omega_0 n + \theta)$$

$$x(n) = \frac{A}{2}e^{j(\omega_0 n+\theta)} + \frac{A}{2}e^{-j(\omega_0 n+\theta)}$$

And the system frequency response is

$$H(e^{j\omega}) = |H(e^{j\omega})|e^{j\angle H(e^{j\omega})}$$

$$y(n) = A|H(e^{j\omega_0})|\cos(\omega_0(n-\tau_p(\omega_0)) + \theta)$$

Where

$$\tau_p(\omega_0) = -\frac{\angle H(e^{j\omega_0})}{\omega_0}$$

is the time delay experienced by the single-frequency signal with frequency ω_0 when it passes through the system. Thus $\tau_p(\omega_0)$ is the system phase delay for a signal with frequency.

Group delay: When the input signal contains many sinusoidal components with different frequencies that are not harmonically related, each component will go through different phase delays when processed by a frequency selective LTI system, and the signal delay is determined using a different parameter called the group delay, as defined below:

$$\tau_g(\omega) = -\frac{d\angle H(e^{j\omega})}{d\omega}$$

The time delay $\tau_g(\omega)$ is called the group delay or envelope delay.

9. Determine the output of the system for each of the following inputs:

$(a)\cos(\pi/2n)$ $(b)\sin(7\pi/2n+\pi/4)$

Solution:

a) The input signal $x(n) = \cos(\pi/2n)$ has a frequency $\omega_0 = \pi/2$. Hence, the response of the system to $x(n)$ is

$$y(n) = |H(e^{j\omega_0})|\cos(\omega_0(n-\tau_g(\omega_0)))$$

$$= |H(e^{j\pi/2})|\cos\left(\frac{\pi}{2}(n-\tau_g(\pi/2))\right)$$

$$y(n) = 2\cos\left(\frac{\pi}{2}(n-2)\right)$$

$$y(n) = 2\cos\left(\frac{\pi}{2}n-\pi\right)$$

b) Consider the given input signal

$$x(n) = \sin\left(\frac{7\pi}{2}n+\frac{\pi}{4}\right)$$

$$x(n) = \sin\left(\frac{\pi}{2}n+\frac{\pi}{4}\right)$$

The input signal $x(n)$ has a frequency $\omega_0 = \pi/2$. Hence, the response of the system to $x(n)$ is

$$y(n) = |H(e^{j\omega_0})|\sin(\omega_0(n-\tau_g(\omega_0)) + \theta)$$

$$= |H(e^{j\pi/2})|\sin\left(\frac{7\pi}{2}(n-\tau_g(\pi/2)) + \frac{\pi}{4}\right)$$

$$y(n) = 2\sin\left(\frac{7\pi}{2}(n-2)+\frac{\pi}{4}\right)$$

$$y(n) = 2\sin\left(\frac{7\pi}{2}n-7\pi+\frac{\pi}{4}\right)$$

$$= 2\sin\left(\frac{7\pi}{2}n - \pi + \frac{\pi}{4}\right)$$

$$y(n) = 2\sin\left(\frac{7\pi}{2}n - \frac{3\pi}{4}\right)$$

Energy Spectral Density (ESD)

Parseval's theorem relates the total signal energy in a signal x(n) to its Fourier transform through

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} \left|X\left(e^{j\omega}\right)\right|^2 d\omega$$

$$= \int_{-1/2}^{1/2} \left|X\left(e^{j\omega}\right)\right|^2 df$$

Relationship between input and output ESDs of an LTI system:

Consider an LTI system with frequency response $H\left(e^{j\omega}\right)$, input y(n). If x(n) and y(n) are energy signals, then their ESDs are $\psi_x\left(e^{j\omega}\right) = \left|X\left(e^{j\omega}\right)\right|^2$ and $\psi_y\left(e^{j\omega}\right) = \left|y\left(e^{j\omega}\right)\right|^2$

$$Y\left(e^{j\omega}\right) = H\left(e^{j\omega}\right)X\left(e^{j\omega}\right)$$

It follows that $\left|Y\left(e^{j\omega}\right)\right|^2 = \left|H\left(e^{j\omega}\right)X\left(e^{j\omega}\right)\right|^2$

$$\left|Y\left(e^{j\omega}\right)\right|^2 = \left|H\left(e^{j\omega}\right)\right|^2 \left|X\left(e^{j\omega}\right)\right|^2$$

Relation of ESD to autocorrelation:

The autocorrelation function $R_{xx}(\tau)$ of a real energy signal is defined as

$$R_{xx}(m) = \sum_{n=-\infty}^{\infty} x(n)x(n-m)$$

$$R_{xx}(m) = x(m) * x(-m)$$

Taking the Fourier transform of the above equation, we have

$$F\left[R_{xx}(m)\right] = X\left(e^{j\omega}\right)X\left(e^{-j\omega}\right)$$

$$= X\left(e^{j\omega}\right)X^*\left(e^{j\omega}\right)$$

$$F\left[R_{xx}(m)\right] = \left|X\left(e^{j\omega}\right)\right|^2$$

$$F\left[R_{xx}(m)\right] = \Psi_x\left(e^{j\omega}\right)$$

$$R_{xx}(m) \longleftrightarrow \Psi_x\left(e^{j\omega}\right)$$

Thus, the autocorrelation function R_{xx} (m) and ESD make a Fourier transform pair.

Power Spectral Density (PSD):

PSD has the same relation to power signals as ESD has to energy signals. The expression of PSD may be obtained by assuming the power signal as a limiting case of an energy signal.

The average power P_x of the signal x(n).

$$P_x = \lim_{N\to\infty} \frac{1}{2N+1}\sum_{n=-N}^{N} |x(n)|^2$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} \lim_{N\to\infty} \frac{\left|X_N\left(e^{j\omega}\right)\right|^2}{2N+1} d\omega$$

$$P_x = \frac{1}{2\pi}\int_{-\pi}^{x} G_x\left(e^{j\omega}\right)d\omega$$

Where

$$G_x\left(e^{j\omega}\right) = \lim_{N\to\infty} \frac{\left|X_N\left(e^{j\omega}\right)\right|^2}{2N+1}$$

is the PSD. The units of PSD depend on the units of the underlying signal x(n).

Common mistake:

Students usually use DFT formula instead of DTFT. Take care of it.

Relationship between input and output PSDs of an LTI system:

Consider an LTI system with frequency response $H(e^{j}\omega)$, input $x_N(n)$, and output $y_N(n)$. If x(n) and y(n) are power signals, then their PSDs are

$$G_x\left(e^{j\omega}\right) = \lim_{N\to\infty} \frac{\left|X_N\left(e^{j\omega}\right)\right|^2}{2N+1}$$

And

$$G_y\left(e^{j\omega}\right) = \lim_{N\to\infty} \frac{\left|Y_N\left(e^{j\omega}\right)\right|^2}{2N+1},$$

Respectively, since we know that

$$Y_N\left(e^{j\omega}\right) = H\left(e^{j\omega}\right)X_N\left(e^{j\omega}\right)$$

If follows that

$$\left|Y_N\left(e^{j\omega}\right)\right|^2 = \left|H\left(e^{j\omega}\right)X_N\left(e^{j\omega}\right)\right|^2$$

$$\left|Y_N\left(e^{j\omega}\right)\right|^2 = \left|H\left(e^{j\omega}\right)\right|^2\left|X_N\left(e^{j\omega}\right)\right|^2$$

$$\lim_{N\to\infty}\frac{\left|Y_N\left(e^{j\omega}\right)\right|^2}{2N+1} = \left|H\left(e^{j\omega}\right)\right|^2 \lim_{N\to\infty}\frac{\left|X_N\left(e^{j\omega}\right)\right|^2}{2N+1}$$

$$G_y\left(e^{j\omega}\right) = \left|H\left(e^{j\omega}\right)\right|^2 G_x\left(e^{j\omega}\right)$$

Relation of PSD to autocorrelation:

The autocorrelation function $R_{xx}(\tau)$ of a power signal is defined as

$$R_{xx}(m) = \lim_{N\to\infty}\frac{1}{2N+1}\sum_{n=-N}^{N} x(n)\,x(n-m)$$

$$= \lim_{N\to\infty}\frac{1}{2N+1}\sum_{n=-\infty}^{N} x_N(n)\,x_N(n-m)$$

$$R_{xx}(m) = \lim_{N\to\infty}\frac{1}{2N+1}\left[x_N(m) * x_N(-m)\right]$$

Taking the Fourier transform of the above equation, we obtain

$$F\left[R_{xx}(m)\right] = \lim_{N\to\infty}\frac{1}{2N+1}X_N\left(e^{j\omega}\right)X_N\left(e^{-j\omega}\right)$$

$$= \lim_{N\to\infty}\frac{1}{2N+1}X_N\left(e^{j\omega}\right)X_N^*\left(e^{j\omega}\right)$$

$$F\left[R_{xx}(m)\right] = \lim_{N\to\infty}\frac{1}{2N+1}\left|X_N\left(e^{j\omega}\right)\right|^2$$

$$F\left[R_{xx}(m)\right] = G_x\left(e^{j\omega}\right)$$

$$R_{xx}(m) \longleftrightarrow G_x\left(e^{j\omega}\right)$$

Thus, the autocorrelation function R_{xx} (m) and PSD makes a Fourier transform pair.

Test Your Understanding

1. Given $x[n] \overset{DTFT}{\longleftrightarrow} X\left(e^{j\omega}\right)$ perform the following calculation without finding DTFT

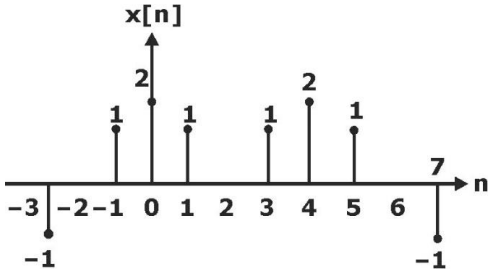


Fig. 6.8

(A) $X(e^{j\omega})$

(B) $\angle X\left(e^{j\omega}\right)$

(C) $\int_{-x}^{x} X\left(e^{j\omega}\right)d\omega$

(D) $\int_{x}^{x}\left|X\left(e^{j\omega}\right)\right|^2$

Solution:

(A) 6

(B) -2ω

(C) 4π

(D) 28π

Cross correlation function:

CCF of two signals x[n] and y[n] are defined as

$$r_{xx}\{k\} = \sum_{n=-\infty}^{\infty} x[n]\,x[n-k] = \sum_{n=-\infty}^{\infty} x[n+k]\,y[n]$$

$$r_{xx}\{k\} = \sum_{n=-\infty}^{\infty} y[n]\,x[n-k] = \sum_{n=-\infty}^{\infty} x[n+k]\,x[n]$$

10. Find the CCF of

$$x[n] = \{1,\underset{\uparrow}{3},2,1\} \;\&\; y[n] = \{4,3,\underset{\uparrow}{1},2\}$$

Solution: $\therefore r_{xx}\{k\} = \sum_{n=-\infty}^{\infty} x[n]\,y[n-k]$

n	-2	-1	0	1	2	3	4	5
X[n]		1	3	2	1			
Y[n]	4	3	1	2				
Y[n+1]	3	1	2					
Y[n+2]	1	2						
Y[n-1]		4	3	1	2			
Y[n-2]			4	3	1	2		
Y[n-3]				4	3	1	2	
Y[n-4]					4	3	1	2

Table 6.1

$$k = -2 \Rightarrow r_{xy}\{-2\} \Rightarrow \sum_{n=-2}^{2} x[n]y[n+2] = 2$$

$$k = -1 \Rightarrow r_{xy}\{-1\} \Rightarrow \sum_{n=-2}^{2} x[n]\,y[n+1] = 1+6 = 7$$

$$k = 0 \Rightarrow r_{xy}\{0\} \Rightarrow \sum_{n=-2}^{2} x[n]\,y[n-0] = 10$$

$$k = 1 \Rightarrow r_{xy}\{1\} \Rightarrow \sum_{n=-2}^{2} x[n]\,y[n-1] = 17$$

Similarly

$$r_{xy}\{2\} = 19; r_{xy}\{3\} = 11; r_{xy}\{4\} = 4$$

If x[n] and y[n] are power signals

$$r_{xy} = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} x[n]\,y[n-k]$$

$$= \lim_{n\to N} \frac{1}{2N+1} \sum_{n=-N}^{N} x[n+k]\,y[n]$$

	Continuous-Time		Discrete-Time	
	Time domain	Frequency domain	Time domain	Frequency domain
Fourier Series	Continuous-time, Periodic in time	Discrete frequency, aperiodic in frequency	Discrete-time, Periodic in time	Discrete frequency, periodic in frequency
Fourier Transform	Continuous-time, Aperiodic in time	Continuous frequency, aperiodic in frequency	Discrete-time, aperiodic in time	Continuous frequency, periodic in frequency

Table 6.2

Exercise

1. The discrete time Fourier transform (DTFT) of $x[n] = 2(3)^n u[-n]$ is equal to

(A) $\dfrac{2}{1-1/3e^{j\Omega}}$

(B) $\dfrac{2}{1+1/3e^{j\Omega}}$

(C) $2\left(\dfrac{1+1/3e^{j\Omega}}{1-1/3e^{j\Omega}}\right)$

(D) $2\left(\dfrac{1-1/3e^{j\Omega}}{1+1/3e^{j\Omega}}\right)$

2. A discrete-time signal $x[n]$ has Fourier transform $X(e^{j\omega})$. Match list–I (Signal) with list–II (Fourier transform) and select the correct answer using the correct answer using the code given below the lists:

List – I	**List – II**
A. $x[-n]$	1. $X^*(e^{-j\omega})$
B. $nx[n]$	2. $X(e^{-j\omega})$
C. $x^*[n]$	3. $e^{-j\omega}X(e^{j\omega})$
D. $x[n-1]$	4. $j\dfrac{d}{d\omega}X(e^{j\omega})$

Codes:

	A	B	C	D
(A)	1	3	2	4
(B)	2	4	1	3
(C)	1	4	2	3
(D)	2	3	1	4

3. Let $x(n) \Leftrightarrow X(e^{j\omega})$ be the discrete time Fourier transform pair.

If $x(n) = 3\delta(n) + \delta(n-1) - 4\delta(n-2) + 6\delta(n-3) + 2\delta(n-4)$.

Then the value of $\sum_{-\infty}^{\infty} n^2 x^2(n)$ is

(A) 906 (B) 66

(C) 132 (D) 453

4. Let $x(n) = 2\delta(n+2) - \delta(n+1) + \delta(n) - \delta(n-1) + 2\delta(n-2)$ with its Fourier transform $X(\omega)$. Then the value of $X(\omega)$ at $\omega = \pi$ is

(A) 7

(B) 5

(C) 6π

(D) -5

5. Let $x(n) = (0.5)^n u(n)$. The DTFT of $x(n)$ is

(A) $\dfrac{e^{j\omega}}{e^{j\omega} + 0.5}$

(B) $\dfrac{1}{1 - 0.5e^{-j\omega}}$

(C) $1 + 0.5e^{-j\omega}$

(D) $\dfrac{1}{1 - 0.5e^{j\omega}}$

Exercise Level - 1

1. Solution: (A)

If $a^n u[n] \xrightarrow{DTFT} \dfrac{1}{1 - ae^{-j\Omega}}$

then $a^{-n} u[-n] \xrightarrow{DTFT} \dfrac{1}{1 - ae^{j\Omega}}$

$x[n] = 2(3)^n u[-n]$

or $x[n] = 2\left(\dfrac{1}{3}\right)^{-n} u[-n]$

So, $2\left(\dfrac{1}{3}\right)^{-n} u[-n] \xrightarrow{DTFT} \dfrac{2}{1 - \dfrac{1}{3}e^{j\Omega}}$

2. Solution: (B)

Properties of Fourier transform:

$x[-n] \xleftrightarrow{F.T.} X\left(e^{-j\omega}\right)$

$nx[n] \xleftrightarrow{F.T.} j\dfrac{d}{d\omega} X\left(e^{j\omega}\right)$

$x^*[n] \xleftrightarrow{F.T.} X^*\left(e^{-j\omega}\right)$

$x[n-1] \xleftrightarrow{F.T.} e^{-j\omega} X\left(e^{j\omega}\right)$

Exercise Level - 2

1. Solution: (D)

$x(n) = 3 + \cos\left(\pi n + \dfrac{\pi}{3}\right) \rightarrow$

$\boxed{H\left(e^{j\omega}\right) = \dfrac{e^{j\omega}}{e^{j\omega} - 1/2}} \rightarrow y[n]$

$h[n] = \left(\dfrac{1}{2}\right)^n u(n)$

$x[n] = 3 + \cos\left(\pi n + \dfrac{\pi}{3}\right) = x_1[n] + x_2[n]$

From the principle of superposition,

$y[n] = y_1[n] + y_2[n]$

DTFT of h(n)

$H\left[e^{j\omega}\right] = \dfrac{e^{j\omega}}{e^{j\omega} - \dfrac{1}{2}}$ ...(i)

Now

$y_1[n] = \left.\dfrac{e^{j\omega}}{e^{j\omega} - 1/2}\right|_{\omega=0} \times 3$

$= \dfrac{1}{1/2} \times 3 = 6$ ...(ii)

and $y_2[n] = \left.\dfrac{e^{j\omega}}{e^{j\omega} - 1/2}\right|_{\omega=\pi} \cdot \cos\left(\pi n + \dfrac{\pi}{3}\right)$

$= \dfrac{2}{3}\cos\left(\pi n + \dfrac{\pi}{3}\right)$ ...(iii)

From equation (ii) and (iii)

Now

$y(n) = y_1[n] + y_2[n]$

$= 6 + \dfrac{2}{3}\cos\left(\pi n + \dfrac{\pi}{3}\right)$

2. Solution: (A)

Given, $h[n] = \delta[n] + \delta[n-1]$

Taking Fourier transform

$H(\Omega) = 1 + e^{-j\Omega}$

$$\Rightarrow H(\Omega) = 2e^{-j\Omega/2}\left(\frac{e^{j\Omega/2} + e^{-j\Omega/2}}{2}\right)$$

$$\Rightarrow H(\Omega) = 2\cos\left(\frac{\Omega}{2}\right)e^{-\frac{j\Omega}{2}}$$

$$\Rightarrow H(\Omega) = 2\cos\left(\frac{\Omega}{2}\right)\angle -\Omega/2$$

3. Solution: (A)

Given: x[n] and h[n] are 16 point sequences

and $y[n] = x[n] \otimes h[n]$ ...(i)

Also, $z[n] = IDFT[X(k) \cdot H(k)]$

Where, $x[n] \xleftrightarrow[16\text{ point}]{DFT} X(k)$

$h[n] \xleftrightarrow[16\text{ point}]{DFT} H(k)$

From circular convolution property of DFT.

$x[n] \textcircled{N}\ h[n] \xleftrightarrow{DFT} X(k) \cdot H(k)$

so, $z[n] = x[n] \textcircled{N}\ h[n]$ ...(ii)

We have to find the value of k, for which

$y(k) = z(k)$

i.e., value n, for which linear and circular convolutions of 16-point sequences x[n] and h[n] are the same.

From definitions of linear convolution,

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k]$$

From definitions of circular convolution,

$$z[n] = \sum_{k=0}^{N-1} x(k) \cdot h[(n-k)]_N$$

where, N=16

Assuming two 16 point sequences as,

$x[n] = \{\underset{\uparrow}{1}, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16\}$

$h[n] = \{\underset{\uparrow}{a}, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p\}$

So, linear convolution y[n] is,

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$$y[0] = \sum_{k=-\infty}^{\infty} x[k]h[-k]$$

$$x[k] = \begin{Bmatrix} \underset{\uparrow}{1}, 2, 3, 4, 5, 6, 7, 8, 9, 10, \\ 11, 12, 13, 14, 15, 16 \end{Bmatrix}$$

$h[-k] = \{p, o, n, m, l, k, j, i, h, g, f, e, d, c, b, \underset{\uparrow}{a}\}$

$\therefore \quad y[0] = (1 \times a) + 0 + 0 + 0 + 0 + 0 + 0$
$+0 + 0 + 0 + 0 + 0 + 0 + 0 + 0 + 0$

$y[0] = a$ ...(i)

From 16-point circular convolution,

$$z[n] = \sum_{K=0}^{15} x(k) \cdot h[(n-k)]_{16}$$

$$z[0] = \sum_{K=0}^{15} x(k) \cdot h[(-k)]_{16}$$

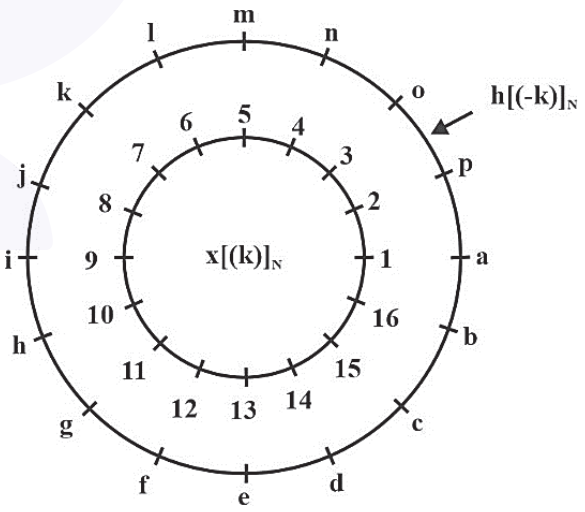


Fig. 6.9

Thus

$z(0) = (a \times 1) + (b \times 2) + (c \times 3) + (d \times 4)$
$+(e \times 5) + (f \times 6) + (g \times 7) + (h \times 8)$
$+(i \times 9) + (j \times 10) + (k \times 11) + (l \times 12)$
$+(m \times 13) + (n \times 14) + (o \times 15) + (p \times 16)$ ...(ii)

From equations (i) and (ii)

$z(0) \neq y(0)$

So, z(k)=y(k) is not satisfied for k=0 and hence options (A), (B) and (D) are wrong.

Now we are choosing option (C) and verifying it as follows,

Linear convolution, $y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k]$

Taking $n = 15, y[15] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[15-k]$

To find h[15-k], shifting sequence h(-k) 15 times towards the right-hand side.

So, $x[k] = \left\{ \begin{matrix} \underset{\uparrow}{1}, 2, 3, 4, 5, 6, 7, 8, 9, 10, \\ 11, 12, 13, 14, 15, 16 \end{matrix} \right\}$

$h[-k] = \{p, o, n, m, l, k, j, i, h, g, f, e, d, c, b, \underset{\uparrow}{a}\}$

$h[15-k] = \{\underset{\uparrow}{p}, o, n, m, l, k, j, i,$

$h, g, f, e, d, c, b, a\}$

$\therefore y(15) = p + 2o + 3n + 4m + 5l + 6k$

$+7j + 8i + 9h + 10g + 11f + 12e$

$+13d + 14c + 15b + 16a$ ...(*iii*)

Circular convolution,

$z[n] = \sum_{k=0}^{15} x(k) \cdot h[(n-k)]_{16}$

$z[15] = \sum_{k=0}^{15} x(k) \cdot h[(15-k)]_{16}$

To find $h\left[\left(15-k\right)\right]_{16}$, rotating $h\left[\left(-k\right)\right]_{16}$ in an anticlockwise direction 15 times. So,

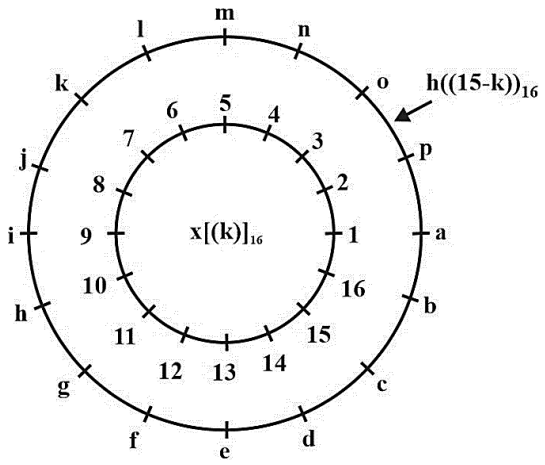


Fig. 6.10

$\therefore z(15) = p + 2o + 3n + 4m + 5l + 6k$

$+7j + 8i + 9h + 10g + 11f + 12e$

$+13d + 14c + 15b + 16a$ ....(iV)

From equations (iii) and (iv)

$y\left(15\right) = z\left(15\right)$

Hence, $y\left(k\right) = z\left(k\right)$ for $k = 15$ is satisfied

Hence, the correct option is (C).